

Scalable Virtual Reality Global Clinical Immersion for Culturally Responsive Engineering Design Skills

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Scalable Global Clinical Immersion Virtual Platform for Culturally Responsive Engineering Design Skills

Introduction:

Global healthcare immersion is challenging for undergraduate biomedical engineering (BME) students due to travel, costs, and logistics. Yet ABET now requires that undergraduates develop the “ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.” Experiences abroad expose students to diverse clinical settings, constraints, and cultural perspectives, boosting problem-solving skills for equitable healthcare solutions. However, many programs rely on short-term extracurriculars, internships, or small capstone cohorts, making large-scale foreign immersion infeasible. To support critical skills in identifying and addressing unmet clinical needs [1-4], clinical immersion in previously inaccessible environments allows BMEs to engage with daily operations and translate unmet needs into human-centric design [5, 6]. ABET highlights unmet clinical needs finding in its “Student Outcome 2: ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety and welfare, as well as global, cultural, social, environmental, and economic factors” [7]. Growing BME cohorts and limited program capacity [8, 9] exclude many students, prompting a shift toward virtual reality (VR) [10, 11] and digital clinical immersion platforms [12, 13] to broaden accessibility.

ABET’s Student Outcome 2 emphasizes that identifying unmet clinical needs must include global healthcare settings, not just local communities. Many undergraduate BME programs offer global engineering experiences through extracurricular programs and international internships, collaborating with organizations like the International Federation for Medical and Biological Engineering [14], Engineering World Health [15], and Engineers Without Borders [16]. Some programs incorporate capstone projects [17], where students work with clinical partners in developing countries. However, these capstones lack direct student immersion in foreign healthcare settings before solution development. Internships and summer programs allow in-person observation but face high costs, housing challenges, and limited student capacity. Many focus on basic medical equipment repair or assessing prototype devices for global clinical partners, rather than engineering new solutions. Similar to clinical immersion programs, they serve only a small number of students, primarily in summer, creating accessibility and scalability constraints. To address accessibility and year-round availability, the Department of Biomedical Engineering at the University of California Irvine (UCI) developed an online platform enabling virtual clinical immersion. Cataract surgeries recorded in the United States, Vietnam, and Paraguay—via 360-degree and head-mounted cameras—immerse students in differing economic and cultural contexts. These dual perspectives offer an immersive 360-degree operating room view combined with a detailed, first-person vantage that can be accessed via computers, VR headsets, and other devices. Supplementary interviews, short lectures, and a questionnaire

reinforce learning on institutional values, regulations, and treatment approaches. Both qualitative and quantitative data measure the platform's effectiveness for teaching global considerations in unmet clinical needs. In Winter 2025, 121 BME undergraduates enrolled (UCI IRB Exempt #5204 for filming; #904 for assessment) and 113 completed the survey, offering qualitative and quantitative insights into global healthcare considerations.

Methods:

For the 2024 academic year, we fully redesigned a prior elective course from 2022 and 2023, "Addressing Unmet Clinical Needs." This revised one-unit asynchronous online class enables students across all UC institutions to engage in virtual clinical immersion and needs finding. It is now a prerequisite for the UCI BME senior capstone program, enhancing students' ability to assess and address unmet needs through prototyping engineered solutions.

Unmet Clinical Needs Finding Course Structure and Learning Outcomes

Students accessed course materials via Canvas LMS (Salt Lake City, UT), viewing weekly 10–15-minute lectures followed by quizzes (Table 1 in Appendix A). These lectures supplemented virtual surgical procedures and interviews, which varied in length by procedure. Cataract surgeries lasted 15 minutes in the U.S. but up to 40 minutes globally due to simultaneous, back-to-back procedures. More complex surgeries, like deep brain stimulation or transcatheter aortic valve replacement, ranged from 30 minutes to an hour. Lecture recordings guided students through the course learning outcomes (Appendix A), namely in unmet clinical needs identification with consideration of non-technical aspects of medical device development. Supplemental materials included the Biodesign textbook [18] (<https://ebiodesign.org/>), YouTube videos, peer-reviewed articles, and other scholarly works. Assessments and lecture topics aimed to: (1) Identify lecture objectives (e.g., Global Unmet Needs Considerations, Figure 1); (2) Practice objectives through group interactions in the virtual clinical immersion platform; and (3) Demonstrate competency via quizzes, an innovation notebook, and a final group report.

Case Study: Diabetes Clinic, Paraguay Slide 6	Global Needs Considerations Slide 10
• An interesting finding while performing cataracts diagnoses in rural areas was the ability to diagnose diabetes through the eye... • As a result, Fundación Vision in Paraguay now has a Diabetes Clinic. • How and Why? • https://www.iapb.org/learn/vision-atlas/causes-of-vision-loss/diabetic-retinopathy/	• Identify how procedures and technologies in these scenarios consider: • The Problem: are we reaching <i>more people or quality of care</i>? • The environment • The government/regulatory framework – see orphanage video! • Available treatments or technologies • The disease burden • Patient population

Figure 1: Sample accompanying learning module lecture slides on global engineering design considerations for use with the global virtual clinical immersion environments.

Virtual Clinical Immersion Platform for Global Medical Procedures

The virtual clinical immersion platform, initially developed with procedures from UCI Medical Center and Children's Hospital of Orange County (CHOC) (IRB Exempt No. 5364, No. 220571)

[10, 11], was expanded to include hospitals in Vietnam and Paraguay (UCI IRB Exempt No. 5204). Using 360-degree stereoscopic and head-mounted cameras, procedures were recorded to provide an immersive operating room view and first-person perspective. Recordings were available in multiple formats. Students with VR headsets accessed 3D viewing, while those using traditional computers experienced a 360-degree panoramic 2D perspective, offering greater immersion than standard videos. The virtual environment, built in Unreal Engine (Epic Games, Cary, NC), supported real-time voice chat and screen sharing, enabling collaborative team viewing if students wanted to meet in groups within the virtual environment. Students with compatibility issues could access all the same videos via a YouTube channel and website (<https://vr.uci.design/>) with interactive 360-degree panning. Supplementary content included healthcare professional interviews, lecture recordings, and online readings on global design principles, unmet needs evaluation, and regulatory strategies (Table 1).

Cataract procedures were recorded in each location, including standard and femtosecond laser-assisted surgeries at UCI Medical Center and standard procedures in Vietnam and Paraguay. U.S. procedures used dedicated operating rooms, while global procedures were performed in shared spaces, reflecting economic and logistical differences. Cataract surgery, performed >20 million times annually [19], was ideal for comparison. All locations followed the standard phacoemulsification technique with intraocular lens implantation [20]. In the U.S., femtosecond laser-assisted procedures were an option for privately insured patients, costing \$245–\$22,000 per Quality Adjusted Life Year [21]. Due to high equipment costs (\$400,000–\$500,000) and annual maintenance (\$25,000–\$50,000) [22], this technology was unavailable in Vietnam and Paraguay, where lower-cost care relies on non-profits. Students engaged with VR experiences, lectures, and interviews, then completed a survey (described below) assessing their ability to recognize differences in values, regulations, and treatment approaches across locations.

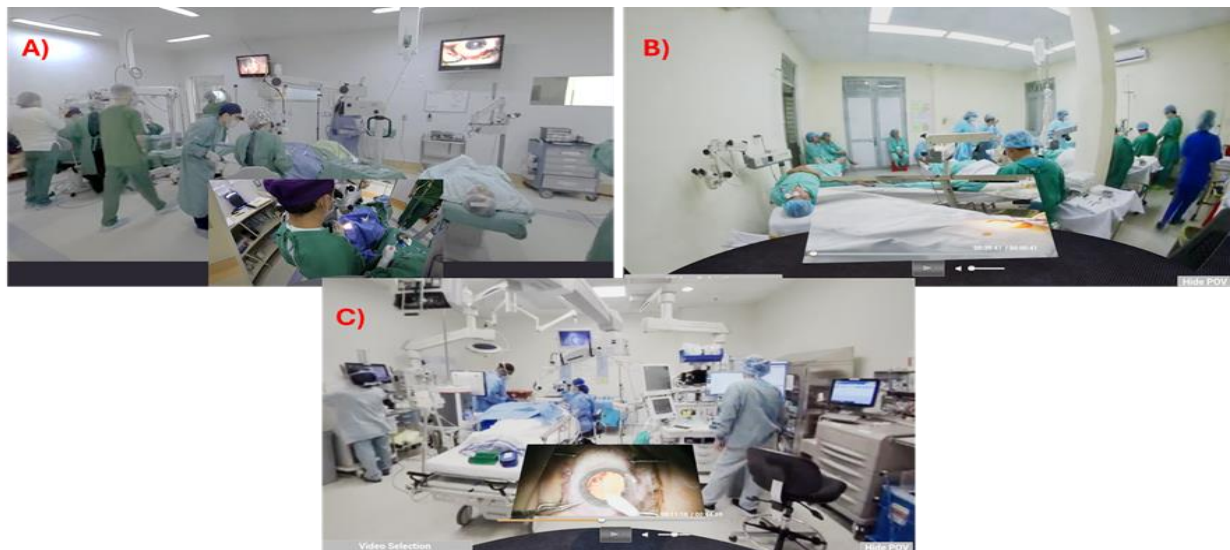


Figure 2: VR clinical immersion environments of a standard cataract surgery performed in A) Paraguay, B) Vietnam, and C) the United States.

Student Surveys and Assessments

To evaluate the feasibility and utility of the virtual clinical immersion platform for unmet clinical needs finding and understanding global healthcare differences, students were asked: Were you able to identify differences in how healthcare is performed across Vietnam, Paraguay, and the U.S.? A student survey (Appendix B) assessed students' ability to recognize differences in standard cataract surgery (UCI IRB Exempt No. 904). A mixed-modality approach using the following quantitative questions and qualitative questions was then utilized to address the research question.

Results:

In Winter 2025, 121 students enrolled in the course, with 113 completing the survey. As shown in Figure 3, 83.19% successfully used the virtual platform, and 62.83% reported increased immersion, presence, or embodiment, enhancing their perception of the operating room environment.

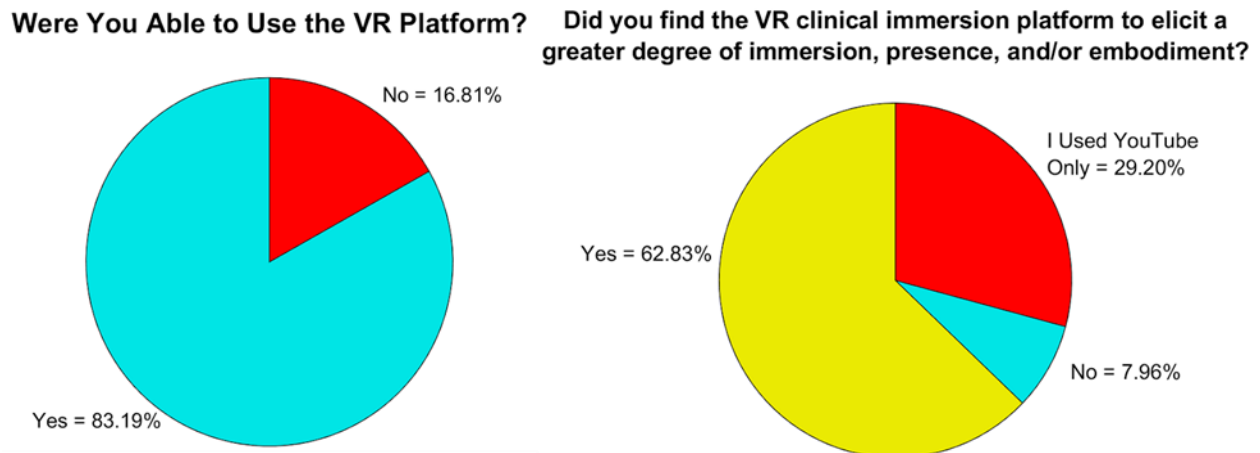


Figure 3: Responses to the questions 1) Were you able to use the VR platform and 2) Did you find the VR clinical immersion platform to elicit a greater degree of immersion, presence, and/or embodiment?

Likert-scale responses were generally positive (Figure 4). Of the 112 students who responded to these items, 65.49% indicated that the clinical immersion videos substantially enhanced their ability to examine how healthcare is performed, and 64.6% reported recognizing differences in clinical operations across countries through the cataract procedure videos.

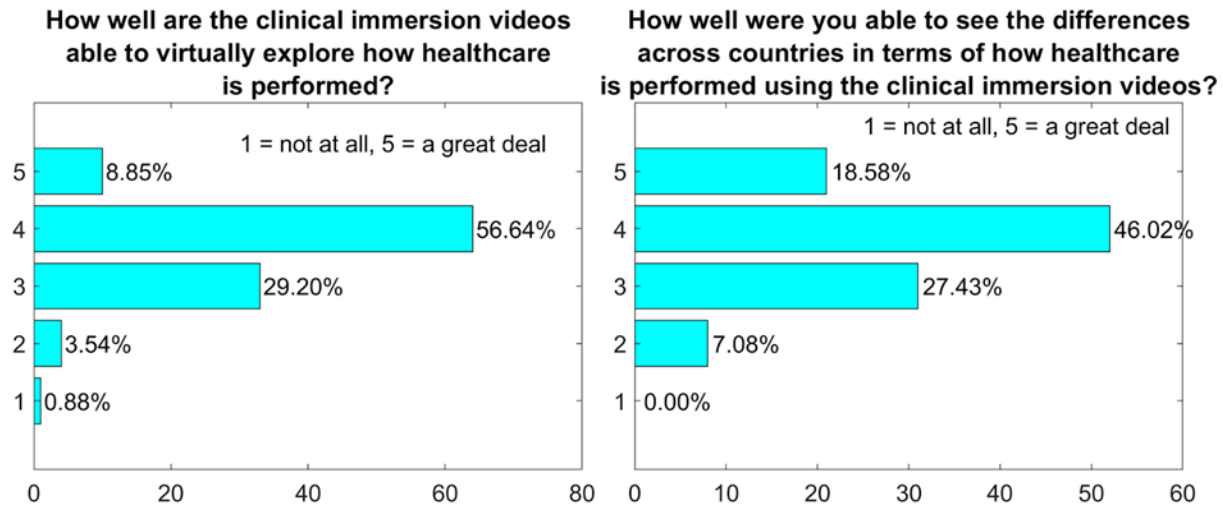


Figure 4: Likert responses to the questions 1) how well are the clinical immersion videos able to virtually explore how healthcare is performed? 2) how well were you able to see the differences across countries (e.g. Vietnam, Paraguay, United States) in terms of how healthcare is performed using the clinical immersion videos?

In addition to the quantitative questions listed above, qualitative analysis was also performed to identify how students were able to learn or if there were any barriers to learning with the inclusion of Question 5: *“Please provide some comments on the global (Vietnam and Paraguay) and local (CHOC and UCI Medical Center) clinical immersion videos you’ve experienced thus far. What have you learned?”* A thematic analysis of 94 survey responses was subsequently performed to elucidate learning outcomes and barriers following the virtual clinical immersion experiences and supplementary learning modules. The analysis identified five major themes: 1) recognition of resource disparities, 2) insights into cultural and systemic factors, 3) enhanced understanding of cataracts and other procedures, 4) understanding the importance of collaboration for successful medical procedures, and 5) connections and applications in biomedical engineering and innovation.

In the first theme, students noted stark differences in technology, operating room conditions, and infrastructure between Vietnam, Paraguay, and the U.S (UCI Medical Center, CHOC). They observed a “lack of advanced medical technology”, “outdated equipment”, and multiple simultaneous surgeries in global settings, while U.S. hospitals had dedicated ORs. For instance, one student noted that:

“Vietnam’s hospital was very cramped and has a lot of patients in one small room. Paraguay had screen technology which Vietnam didn’t seem to have.”

“I noticed that for most of the Vietnam videos, the procedures would take place with multiple people within a room whereas the others generally stuck to one patient per room. When it came to cataract surgery, it also seemed like the ones done in Vietnam required more people and were

more hands-on compared to other versions. Considering these videos, I learned that certain surgeries can be done in an efficient manner, where particular focus could be put on the rate of patient procedures and to what extent resources should be used to maximize speed without costing effectiveness.”

These differences in medical technology, particularly in Vietnam, stem from non-profit hospital goals to provide low or no cost care to more patients. In contrast, U.S. procedures occurred in for-profit hospitals that accept private and public insurance.

The second theme revealed student observations on cultural practices, government roles, and healthcare attitudes. One noted that in Paraguay, “*all people are required to be an organ donor;*” while another highlighted that in Vietnam, “*family members assume significant caregiving roles.*” Students also identified differences in pre and post procedure care.

“Since I am Vietnamese, I have watched more of the Vietnam videos than I have the Paraguay ones, however for both, I noticed the lack of technology in the clinics and overall diagnostic progress. In the US, I am very accustomed to seeing technology used in even small clinics where people would use iPads to track data or record info and machines that take measurements for the user rather than traditional methods. In Vietnam in particular, the videos gave me an interest in the difference between hospitals and clinics here, so I asked some of my aunts who have had experience in Vietnam’s hospitals for some key differences they experienced. Apparently, in Vietnam, patient care differs from the US where nurses do not care for patients, rather it is the patient’s family’s job to do so. In Vietnam, nurses only help doctors, while patient care is delegated to the family.”

The third theme highlighted students' deeper understanding of surgical techniques and the universal need for specialized tools and sterile environments. They were especially struck by the “*delicate*” nature of eye surgeries. One student noted:

“I have learned a lot more about the process of cataract surgery and how delicate the surgeons need to be since they are working so closely with the eye.”

Theme four highlighted the importance of team coordination in medical procedures. Students noted the involvement of “*multiple doctors and nurses*” and that “*there are a lot of people in the room while the procedure is happening.*” They also observed variations in roles and responsibilities across countries:

“I didn’t realize how many people there are in the room while the procedure is happening. ... a bit of a shock, but it was really interesting to me.”

“Seeing how medical teams adapt in different settings has helped me understand the real world needs for biomedical innovations. This experience is crucial for designing solutions that address gaps in care, whether in resource limited environments or advanced hospitals.”

Lastly, students connected virtual clinical observations to potential biomedical engineering improvements, recognizing the need for “*cost-effective and scalable solutions*” to expand patient care. They also identified “*gaps in care*” that could be addressed through engineering innovation:

“I learned here that it is important to see how these devices are used in a medical setting rather than a controlled environment... it allows for gaps and flaws to be rapidly identified.”

“After watching the clinical immersion films, I have seen an interesting point of view on addressing the unmet clinical needs in both local and global contexts. I envisioned creative problem-solving skills fueled by a lack of resources in Vietnam, such as the development of portable and affordable medical devices. This emphasizes how important it is to give durability and accessibility first priority when it comes to solving healthcare issues. The emphasis on cutting-edge technologies and specialized care at CHOC and UCI Medical Center proves how innovation is the only way to meet the unmet clinical needs locally, for example, improving pediatric care or pushing the workflows with electronic health records. As a biomedical engineer, I understand that it is necessary to create solutions that seek the balance among creativity, viability, and sustainability in order to resolve a range of clinical requirements.”

Students reported significant learning outcomes, particularly in understanding global resource disparities, procedural differences, and cultural influences on healthcare. While virtual clinical immersion was widely valued, technical barriers limited its effectiveness. Issues such as pixelation, poor video quality, motion sickness, and insufficient narration or a “*lack of explanation*” hampered learning. Since the platform lacked advanced video compression algorithms, students often had to stream videos at suboptimal resolutions (720p or low-bitrate 1080p) due to bandwidth limitations, particularly on UCI’s campus. This resulted in choppy or laggy visuals, contributing to motion sickness. One student noted: “*Honestly, the videos were great, but I personally couldn’t focus on the VR video because I get motion sickness.*” Similarly, some students noted the lack of visual quality as limiting their full understanding of how different resources impact surgical workflows, while others struggled with “*seeing the actual or close-up procedure*” due to pixelation. This, coupled with a lack of informative narration limited their ability to analyze surgical workflows and resource impact.

“...As my laptop is unable to support high resolution, it was difficult for myself to make sense of the video, especially without proper narration from the surgeon or team...”

While students voiced challenges such as motion sickness and technical issues, many recognized our virtual platform’s potential to immerse them in clinical environments and found the immersive perspective valuable for comparing global versus local approaches.

Discussion:

Survey results confirmed the virtual platform effectively supported clinical immersion, enhancing students' understanding of global healthcare disparities. While 29.20% (N = 33) used YouTube due to MacOS compatibility or storage issues, most found it valuable for analyzing

cross-country procedural differences. Qualitative analysis showed students recognized key healthcare considerations like resource availability, cultural factors, and infrastructure variations, aligning with ABET Student Outcome 2 on designing for diverse healthcare contexts. The platform also fostered collaboration between engineers and healthcare professionals, with some students conducting independent interviews to deepen their understanding of cultural values in healthcare design, reflecting ABET Student Outcome 7: “an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.” To improve usability, updates will resolve MacOS/Linux compatibility issues, enhance video quality, and reduce storage requirements (currently 3.3 GB). Mobile support (iOS, Android) will be prioritized to expand accessibility beyond Windows PCs and VR headsets. Interactive features like real-time quizzes, narrated modules, and engagement alerts will be integrated to address student feedback on boredom, informed by research on cognitive engagement and retention [23, 24]. Strategic partnerships with Engineering World Health, Engineers Without Borders, and for-profit hospitals will expand global procedure repositories. Near-peer mentorship programs [25] will enable international collaboration on capstone projects addressing unmet clinical needs. Future research will assess the educational benefits, intellectual property generation, and formation of startups/non-profits, demonstrating global partnerships’ role in healthcare innovation. If successful, this initiative will provide key insights into the impact of global collaborations in engineering education and healthcare solutions.

Conclusion:

The findings demonstrate the transformative potential of virtual clinical immersion in educating students on global and local healthcare challenges through engineering design principles. Qualitative and quantitative analyses confirm its effectiveness in teaching students to identify and address unmet clinical needs. A major advantage is its scalability and accessibility. By leveraging open-source software, the platform removes barriers such as high costs, limited capacity, visa restrictions, and seasonal availability, allowing unlimited student participation across socioeconomic and educational backgrounds. Integrating virtual immersion with asynchronous coursework, it offers a flexible, self-paced learning environment for exploring global healthcare systems. Future research will track how unmet needs evolve into capstone projects, intellectual property, and startups or non-profits, demonstrating the platform’s long-term impact on healthcare innovation and engineering education.

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Appendix A: Student Learning Outcomes and Lecture Topics of the Course

Student Learning Outcomes of the Course (EAC refers to ABET Student Outcomes 1-7 [7]):

1. Identify medical, technical, and market needs in formulating biomedical engineering solutions to unmet clinical needs (EAC 7).
 1. Students will understand the technical requirements for medical device design.
 2. Students will learn how to use human-centered engineering approaches in medical device design.
 3. Students will evaluate the potential commercialization of solutions to unmet clinical needs through market analyses and business strategies.
2. Employ the various steps in bringing a biomedical product from ideation to invention (EAC 2).
 1. Students will learn how to identify unmet clinical needs in various medical fields.
 2. Students will evaluate unmet clinical needs to develop novel solutions.
 3. Students will learn how to professionally communicate a proposed invented or innovative solution to an unmet clinical need.
3. Identify the realistic constraints of a medical innovation product, with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors (EAC 2).
 1. Students will be able to identify and evaluate global, cultural, social and environmental factors that are important in medical device design.
 2. Students will be able to analyze the economic factors that are considered during medical device design, and how these impact design and the lifecycle of the product.
 3. Students will learn how to identify and apply the regulatory and safety requirements of bringing a medical device to market.
4. Articulate the impacts of the project in a global, economic, environmental and societal context (EAC 4).
 1. Students will learn to integrate regulatory, intellectual property, and global, cultural, environmental, and economic considerations into their solutions for identified clinical needs.
5. Communicate unmet clinical need, concept generation, and concept screening of medical innovation and invention to a wide range of audiences (EAC 3).

1. Students will learn how to communicate solutions of unmet clinical needs to a wide range of academic and non-academic audiences and forums, including thorough technical reports.

Table 1: Learning topics by week in the online course.

Week	Lecture Topic
1	Team Formation and Leadership
2	Identifying Unmet Needs and Engineering Design
3 and 4	Evaluating Unmet Clinical Needs
5	Human Centered Engineering Design
6	FDA and Reimbursement Strategies
7	<i>Individualized Team and Project Performance Assessments and Feedback</i>
8	Intellectual Property Basics
9	Go-To Market Strategy and Commercialization
10	Global Considerations in Unmet Clinical Needs

Appendix B: Survey Questions

1. Are you able to use the VR platform provided?
2. Did you find the VR clinical immersion platform to elicit a greater degree of immersion, presence, and/or embodiment (the feeling that you are actually present in the operating rooms/clinical environment)?
3. On a scale of 1 to 5, how well are the clinical immersion videos able to virtually explore how healthcare is performed (1= not at all, 5 = a great deal)?
4. On a scale of 1 to 5, how well were you able to see the differences across countries (e.g. Vietnam, Paraguay, United States) in terms of how healthcare is performed using the clinical immersion videos? (1 = not at all, 5 = a great deal)
5. Please provide some comments on the global (Vietnam and Paraguay) and local (Children's Hospital of Orange County and UCI Medical Center) clinical immersion videos you've experienced thus far. What have you learned?